



Problem

Euclid's Identity states that for any two positive integers A and B, there exist integers x and y such that: $Ax + By = D$

where D is the Greatest Common Divisor (GCD) of A and B.

Given two positive integers A and B, find: $x \ y \ D = \gcd(A, B)$

such that:

$$Ax + By = D$$

If multiple pairs (x, y) satisfy the equation, choose the pair with the minimum value of: $|x| + |y|$

If there is still more than one pair, choose the pair for which:

$$x \leq y$$

Submissions

Input Format

A B

Leaderboard

Constraints

$$1 \leq A, B \leq 10^9$$

Output Format

$x \ y \ D$ D is $\gcd(A, B)$ $Ax + By = D$ (x, y) satisfies the minimum-selection rules.

Discussions

Sample Input 0

4 6

[Change Theme](#)

Language

Python 3



```
1 # Enter your code here. Read input from STDIN. Print
2 def egcd(a, b):
3     if b == 0:
4         return a, 1, 0
5     d, x, y = egcd(b, a % b)
6     return d, y, x - (a // b) * y
7
8 A, B = map(int, input().split())
9
10 d, x, y = egcd(A, B)
11 best = (x, y)
12
13 for k in range(-2, 3):
14     a = x + k * (B // d)
15     b = y - k * (A // d)
16
17     if abs(a) + abs(b) < abs(best[0]) + abs(best[1]):
18         best = (a, b)
19     elif abs(a) + abs(b) == abs(best[0]) + abs(best[1])
20         best = (a, b)
21
```

Line: 10 Col: 21



Upload Code as File



Test against custom input

Run Code

Submit Code